CHEN Tianyuan

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Forward Deployed Engineer

AI System Delivery & Engineering Lead

LANGUAGES Mandarin (native) · Cantonese (basic conversational) · English (basic conversational) | **CERTIFICATIONS** PMP · CSM · CET-6

WORK AUTHORISATION Hong Kong — IANG (Immigration Arrangements for Non-local Graduates) visa **approved** · 24-month stay · **no employer sponsorship required**

7	50	1	20+ / 6
AI models taken to production	engineers led across 5 teams	round to pass expert review	years engineering / leading

PROFILE

Forward Deployed Engineer with **20+ years of software engineering** and **6+ years leading teams of up to 50**, plus an **MSc in Applied Artificial Intelligence (Hong Kong)**. I turn ambiguous, high-stakes AI ambitions into systems that run in production: scoping with non-technical stakeholders, designing the architecture, training and compressing the models, deploying on-prem or at the edge, then carrying the solution through UAT and expert sign-off.

WHAT I BRING TO AN FDE ROLE

- **Discovery under ambiguity** — turning a one-paragraph brief from a client or research team into a modular scope with defined interfaces and acceptance criteria.
- **End-to-end ownership & coordination** — architecture, model training, deployment automation, testing and handover, while coordinating universities, vendors and client leadership.
- **Hard deployment environments** — air-gapped GPU nodes, offline installation, constrained edge compute, strict audit, data-residency and NDA requirements.
- **AI that ships, and can be explained** — RAG, LLM agents, forecasting and anomaly detection in real use; 3 live technical talks delivered.

CORE SKILLS

Applied AI / ML	PyTorch, LLM fine-tuning, RAG, LangChain, vector databases, model evaluation, compression & quantisation, edge AI, time-series forecasting, anomaly detection.
GPU & Model Serving	NVIDIA H100 / H20 server bring-up, driver & container-runtime deployment, vLLM model serving, air-gapped installation, node / service monitoring, capacity tuning.
Engineering & Data	Python, Java, Spring Boot / Spring MVC / MyBatis, microservices, multithreading, JVM tuning, distributed systems, MySQL (read/write splitting), Redis Cluster, RocketMQ, data pipelines, Docker, Kubernetes, CI/CD.
Delivery & Governance	Requirements discovery, scope control, vendor management, UAT & acceptance evidence, expert-review panels, NDA handling, audit-ready documentation.

PROFESSIONAL EXPERIENCE

Forward Deployed Engineer / Independent Technical Consultant

Mar 2026 – Present | Hong Kong · Mainland China

▸ **National-Level Infrastructure AI Optimisation Platform** (client name withheld under NDA) — 2026 – Present

- Sole delivery owner for the AI layer of a national-level infrastructure programme; converted academic algorithms into **7 production models** covering forecasting, scheduling, optimisation and anomaly detection.
- Ran requirements discovery directly with the client and the research team, decomposing a vague and fast-moving brief into independently deployable modules with defined interfaces, boundaries and acceptance criteria.
- Built a **zero-dependency deployment toolkit** so the platform could be installed, configured and debugged on **offline, resource-constrained field nodes** with no internet access — removing the need for vendor support on site.
- Brought up the programme's **air-gapped GPU inference environment** on NVIDIA H20 nodes: driver and container-runtime deployment, the vLLM serving stack, and node / service monitoring — with no external package source reachable from the cluster.
- Cut model size and inference latency through compression and quantisation to fit on-site hardware limits.
- Authored test plans, acceptance evidence and the technical Q&A pack for expert review; the platform **passed expert review in a single round**.

Independent GPU Inference Lab (self-funded R&D infrastructure) 2026

- Built and operated an on-premise **NVIDIA H100** inference server running **vLLM** to serve a **32B-parameter model**, patching the agent runtime to extend usable context to **64K tokens** — the same stack I now deploy into client environments.

Smart Infrastructure AI Innovation Project — AI Component

- Owned the AI workstream end to end: algorithm review from the literature, feature engineering, training, edge deployment and handover documentation — with no standing engineering team behind me.
- Coordinated the university research team (theory) and external vendors (non-core modules), keeping the critical path in-house and integrating third-party deliverables against a fixed acceptance date.
- Produced the compliance and audit documentation required for a government-funded programme.

AI Transformation for a Cross-Border E-Commerce Business 2025 – 2026

- Diagnosed the manual bottlenecks in the client's operations, then shipped an **LLM + RAG assistant** for product-listing generation, customer service and data analysis, integrated into the existing e-commerce platform.
- Owned the full cycle from diagnosis through production cut-over and staff handover training; significantly reduced manual workload on repetitive listing and after-sales tasks.

AI Knowledge Base & Workflow Assistant for a Tech Startup (WeCom Integration) 2025 – 2026

- Deployed an on-premise **RAG knowledge base** and automated workflow assistant into WeCom, enabling intelligent document retrieval, automated replies and internal process automation for a lean team.

Senior Project Manager / Technical Lead — Chinasoft International (ICSS), Huawei Vmall **Jun 2019 – Oct 2025**

- Delivered and operated the **customer service subsystem of Huawei's Vmall e-commerce platform**, staying hands-on across a Spring Boot / MySQL (read/write splitting) / Redis Cluster / RocketMQ stack.
- Led performance and stability work directly: JVM and SQL tuning, cache penetration and avalanche protection, and capacity design for high-concurrency traffic peaks.
- Managed up to **50 engineers** across multiple development teams, owning delivery quality, QA and HR functions (recruitment, onboarding, training).
- Led the AMS team to **#1 in quality rankings six times in 2021** across Vmall projects; drove continuous improvement for 5 teams, with 3 consistently holding top-5 quality rankings.
- Represented the e-commerce business line in **3 live technical talks**, receiving multiple awards.

EDUCATION

MSc, Applied Artificial Intelligence
Hong Kong Chu Hai College | 2025 – 2026

BSc, Information and Computing Science
Wuhan University of Science and Technology | 2002 – 2006

CERTIFICATIONS

PMP
Project Management Professional

CSM
Certified Scrum Master

CET-6
College English Test, Band 6

AVAILABILITY

Hong Kong work authorisation
IANG visa approved — no employer sponsorship needed.

Based in Hong Kong
Open to client-site work and travel across the Greater Bay Area and APAC.

Comfortable with NDA-bound work
Client and project names withheld where required.

References available upon request. Project details are presented in a desensitised form in line with client NDAs; technical architecture and delivery approach can be discussed in detail during interview.

HOW I RUN A DEPLOYMENT

1 · Frame
Sit with the client and the research team; agree the single outcome that matters and how it will be judged.

2 · Scope
Break the brief into independently deployable modules with owners, interfaces and acceptance criteria.

3 · Prototype
Get a thin end-to-end slice running on the client's real data in days — before hardening anything.

4 · Harden
Quantisation, offline packaging, monitoring and failure modes: the work that decides survival on site.

5 · Hand over
Acceptance evidence, audit documentation, training and expert-review Q&A support.